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Flexible display screen and method for assembling the same, and electronic apparatus

Abstract

A flexible display screen includes a main display portion, a first bending portion, a second bending portion and a corner bending portion. The main display portion includes a first side, an arc-shaped side and a second side that are connected in sequence. The first bending portion is located on a side of the main display portion where the first side is located, and is connected to the first side. The second bending portion is located on a side of the main display portion where the second side is located, and is connected to the second side. The corner bending portion is located on a side of the main display portion where the arc-shaped side is located, and is connected to the arc-shaped side. A plurality of openings are disposed in the corner bending portion, and face a bending direction of the corner bending portion.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
9441828	12/2015	Wu	N/A	F21V 33/0052
10903299	12/2020	Shim	N/A	G06F 1/1601
10939547	12/2020	Lee	N/A	H05K 1/0313
10990133	12/2020	Nakatogawa	N/A	G09G 3/3266
11528812	12/2021	Ahn	N/A	G09F 9/302
11528813	12/2021	Jeong	N/A	G06F 1/1637
11599213	12/2022	Joo	N/A	G06F 3/0446
11659742	12/2022	Won	257/72	H10K 59/1213
11678509	12/2022	Hyun	257/40	G06F 1/1626
11806974	12/2022	Cao	N/A	B32B 3/30
11825684	12/2022	Park	N/A	H10K 59/8791
11922843	12/2023	Wang	N/A	G06F 1/1637
11968789	12/2023	Ahn	N/A	G06F 1/1637
11977417	12/2023	Park	N/A	G06F 1/1637
12019819	12/2023	Joo	N/A	G06F 3/0412
2014/0218971	12/2013	Wu	362/613	F21V 33/0052
2016/0021226	12/2015	Allore et al.	N/A	N/A
2019/0081127	12/2018	Shim	N/A	H10K 77/111
2019/0269011	12/2018	Lee	N/A	H05K 3/0014
2020/0142449	12/2019	Nakatogawa	N/A	H05B 33/02
2020/0170126	12/2019	Ahn	N/A	G09F 9/33
2020/0176696	12/2019	Dai	N/A	N/A
2020/0221588	12/2019	Jeong	N/A	H05K 5/0017

2021/0153363	12/2020	Cao et al.	N/A	N/A
2021/0165454	12/2020	Dong et al.	N/A	N/A
2021/0234120	12/2020	Zhang et al.	N/A	N/A
2021/0318770	12/2020	Joo	N/A	G06F 3/04164
2021/0376038	12/2020	Won	N/A	H10K 77/111
2021/0384465	12/2020	Park	N/A	H10K 50/844
2022/0080697	12/2021	Cao	N/A	B32B 3/02
2022/0123256	12/2021	Hyun	N/A	H10K 50/844
2023/0029926	12/2022	Park	N/A	G06F 1/1656
2023/0106694	12/2022	Ahn	361/807	G06F 1/1637
2023/0111683	12/2022	Wang	345/55	G06F 1/1652
2023/0205347	12/2022	Joo	345/173	G06F 3/0446
2024/0040826	12/2023	Park	N/A	H10K 59/12
2024/0153419	12/2023	Wang	N/A	G09G 3/035
2024/0206080	12/2023	Ahn	N/A	H05K 5/0017

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
105492991	12/2015	CN	N/A
108108062	12/2017	CN	N/A
108615463	12/2017	CN	N/A
108667960	12/2017	CN	N/A
108766977	12/2017	CN	N/A
109032251	12/2017	CN	N/A
110010013	12/2018	CN	N/A
110518039	12/2018	CN	N/A
110767096	12/2019	CN	N/A
110943114	12/2019	CN	N/A
110992828	12/2019	CN	N/A
110992831	12/2019	CN	N/A
111681547	12/2019	CN	N/A

OTHER PUBLICATIONS

Office Action for the Chinese Patent Application No. 202010522838.2 issued by the Chinese Patent Office on Aug. 27, 2021. cited by applicant

Notification to Grant Patent Right for Invention for the Chinese Patent Application No. 202010522838.2 issued by the Chinese Patent Office on Feb. 9, 2022. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a national phase entry under 35 USC 371 of International Patent Application No. PCT/CN 2021/099220 filed on Jun. 9, 2021, which claims priority to Chinese Patent

Application No. 202010522838.2, filed on Jun. 10, 2020, titled "FLEXIBLE DISPLAY SCREEN AND METHOD FOR ASSEMBLING THE SAME, AND ELECTRONIC APPARATUS", which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the technical field of electronic apparatuses, and in particular, to a flexible display screen and a method for assembling the same, and an electronic apparatus.

BACKGROUND

(3) With the rapid development of display technologies, trends of beauty, lightness and thinness of an electronic product gradually emerge, and an attaching of a curved surface of a flexible display screen is rapidly developed.

(4) With the gradual maturity of a curved attaching process in two sides of the flexible display screen, a curved attaching process in four sides of the flexible display screen has gradually begun to develop. However, in the curved attaching process in the four sides of the flexible display screen, it is difficult to realize a corner attaching of the flexible display screen. Especially with an increase of a depth of a corner arc-shaped region, which is attached to a forming die, of the flexible display screen, the flexible display screen is stressed increasingly complicatedly.

SUMMARY

(5) In an aspect, a flexible display screen is provided. The flexible display screen includes a main display portion, a first bending portion, a second bending portion and a corner bending portion.

(6) The main display portion includes a first side, an arc-shaped side and a second side that are connected in sequence. The first bending portion is located on a side of the main display portion where the first side is located, and is connected to the first side. The second bending portion is located on a side of the main display portion where the second side is located, and is connected to the second side. The corner bending portion is located on a side of the main display portion where the arc-shaped side is located, and is connected to the arc-shaped side.

(7) The first bending portion, the corner bending portion and the second bending portion are connected in sequence, and constitute a display region together with the main display portion. A plurality of openings are disposed in the corner bending portion, and face a bending direction of the corner bending portion.

(8) In some embodiments, the corner bending portion has a wiring region and a deformation region that are sequentially disposed along a direction from the corner bending portion toward the main display portion. The deformation region is connected to the arc-shaped side, and the plurality of openings are disposed in the deformation region. At least one first signal line is disposed in the wiring region.

(9) In some embodiments, the plurality of openings are arranged in a plurality of rows in a radial direction of the arc-shaped side, and openings in each row are arranged at intervals along an extending direction of the arc-shaped side.

(10) In some embodiments, along a direction from a center of the deformation region toward the first bending portion and along a direction from the center of the deformation region toward the second bending portion, dimensions of openings in each of at least one row along the radial direction of the arc-shaped side are reduced in sequence; and/or dimensions of openings in each of at least one row along the extending direction of the arc-shaped side are reduced in sequence.

(11) In some embodiments, along a direction from a center of the deformation region toward a side boundary of the deformation region along the radial direction of the arc-shaped side and along a direction from the center of the deformation region toward another side boundary of the deformation region along the radial direction of the arc-shaped side, dimensions, along the radial direction of the arc-shaped side, of openings arranged along the radial direction of the arc-shaped side are reduced in sequence; and/or dimensions, along the extending direction of the arc-shaped side, of openings arranged along the radial direction of the arc-shaped side are reduced in sequence.

(12) In some embodiments, the deformation region includes a middle region and an edge region,

and the edge region surrounds the middle region. A dimension, along the radial direction of the arc-shaped side, of an opening in the plurality of openings located in the middle region is greater than a dimension, along the radial direction of the arc-shaped side, of an opening in the plurality of openings located in the edge region; and/or a dimension, along the extending direction of the arc-shaped side, of an opening in the plurality of openings located in the middle region is greater than a dimension, along the extending direction of the arc-shaped side, of an opening in the plurality of openings located in the edge region.

(13) In some embodiments, dimensions, along the radial direction of the arc-shaped side, of openings in the plurality of openings located in the middle region are substantially same; and/or dimensions, along the radial direction of the arc-shaped side, of openings in the plurality of openings located in the edge region are substantially same.

(14) In some embodiments, a dimension of the deformation region along the radial direction of the arc-shaped side is $L1$, and a minimum distance, along the radial direction of the arc-shaped side, between a boundary of the middle region and a boundary of the deformation region that are close to each other is in a range of $\frac{1}{4} L1$ to $\frac{1}{3} L1$; and/or a dimension of the deformation region along the extending direction of the arc-shaped side is $L2$, and a distance, along the extending direction of the arc-shaped side, between a boundary of the middle region and a boundary of the deformation region that are close to each other is in a range of $\frac{1}{6} L2$ to $\frac{1}{5} L2$.

(15) In some embodiments, along a direction from a center of the deformation region toward the first bending portion and along a direction from the center of the deformation region toward the second bending portion, distances, each of which is a distance between two adjacent openings in openings 4 in each of at least one row, are increased in sequence; and/or along a direction from a center of the deformation region toward a side boundary of the deformation region along the radial direction of the arc-shaped side and along a direction from the center of the deformation region toward another side boundary of the deformation region along the radial direction of the arc-shaped side, distances, each of which is a distance between two adjacent openings in openings arranged along the radial direction of the arc-shaped side, are increased in sequence.

(16) In some embodiments, an opening in the plurality of openings has a strip shape, and a length extending direction of the opening is substantially same as an extending direction of the arc-shaped side; and/or a length extending direction of the opening is substantially tangent to an extending direction of the arc-shaped side.

(17) In some embodiments, the plurality of openings are divided into a plurality of groups that are arranged in a plurality of rows along a radial direction of the arc-shaped side, and groups of openings in each row are arranged at intervals along an extending direction of the arc-shaped side or a tangential direction of the arc-shaped side. A distance between two adjacent openings in each group of openings is less than a distance between two openings that are closest to each other in two adjacent groups of openings.

(18) In some embodiments, an opening in the plurality of openings has a circular shape, an elliptical shape, or a polygonal shape.

(19) In some embodiments, an opening in the plurality of openings is a through hole penetrating through the corner bending portion along a thickness direction of the corner bending portion; or an opening in the plurality of openings is a groove recessed along a thickness direction of the corner bending portion.

(20) In some embodiments, the corner bending portion includes a plurality of sub-pixels. Two adjacent openings are provided with at least one sub-pixel therebetween; or the plurality of openings are divided into a plurality of groups, and two adjacent groups of openings are provided with at least one sub-pixel therebetween.

(21) In some embodiments, the corner bending portion has a wiring region and a deformation region. The plurality of openings are disposed in the deformation region, and at least one first signal line is disposed in the wiring region. The corner bending portion further includes a plurality

of second signal lines. Each sub-pixel is electrically connected to at least one second signal line. The plurality of second signal lines extend in gaps between the plurality of openings, and each second signal line is electrically connected to a first signal line in the at least one first signal line.

(22) In another aspect, a method for assembling a flexible display screen is provided. The flexible display screen is the flexible display screen in any one of the above embodiments. The assembling method includes: connecting a side of the first bending portion, a side of the second bending portion, and a side of the corner bending portion that are away from the main display portion to respective carrier films; attaching the flexible display screen to a forming die, at least part of each of the carrier films protruding from a boundary of the forming die; and pulling the carrier films to attach the flexible display screen to the forming die.

(23) In some embodiments, a carrier film connected to the corner bending portion of the flexible display screen is provided with a plurality of slits arranged at intervals, and the plurality of slits are arranged at intervals along an extending direction of the arc-shaped side of the main display portion of the flexible display screen. Each slit extends along a radial direction of the arc-shaped side, and an end of each slit extends to a side of the carrier film connected to the corner bending portion.

(24) In yet another aspect, an electronic apparatus is provided. The electronic apparatus includes the flexible display screen in any one of the above embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In order to describe technical solutions in the present disclosure more clearly, accompanying drawings to be used in some embodiments of the present disclosure will be introduced briefly below. Obviously, the accompanying drawings to be described below are merely accompanying drawings of some embodiments of the present disclosure, and a person of ordinary skill in the art may obtain other drawings according to these drawings. In addition, the accompanying drawings to be described below may be regarded as schematic diagrams, but are not limitations on an actual size of a product, an actual process of a method and an actual timing of a signal to which the embodiments of the present disclosure relate.
- (2) FIG. 1 is a structural diagram of an electronic apparatus, in accordance with some embodiments;
- (3) FIG. 2 is a structural diagram of a flexible display screen, in accordance with some embodiments;
- (4) FIG. 3 is a structural diagram corresponding to the A region in FIG. 2;
- (5) FIG. 4 is a structural diagram corresponding to the C region in FIG. 3;
- (6) FIG. 5 is another structural diagram corresponding to the C region in FIG. 3;
- (7) FIG. 6 is yet another structural diagram corresponding to the C region in FIG. 3;
- (8) FIG. 7 is a structural diagram corresponding to the D region in FIG. 3;
- (9) FIG. 8 is another structural diagram corresponding to the D region in FIG. 3;
- (10) FIG. 9 is yet another structural diagram corresponding to the D region in FIG. 3;
- (11) FIG. 10 is a diagram showing a distribution of openings corresponding to a deformation region in FIG. 3;
- (12) FIG. 11 is a diagram showing another distribution of openings corresponding to a deformation region in FIG. 3;
- (13) FIG. 12 is yet another structural diagram corresponding to the C region in FIG. 3;
- (14) FIG. 13 is yet another structural diagram corresponding to the D region in FIG. 3;
- (15) FIG. 14 is a diagram showing yet another distribution of openings corresponding to a deformation region in FIG. 3;
- (16) FIG. 15 is a sectional view of openings in a deformation region in FIG. 3 taken along the EE'

direction;

(17) FIG. 16 is another sectional view of openings in a deformation region in FIG. 3 taken along the EE' direction;

(18) FIG. 17 is a diagram showing the shapes of openings, in accordance with some embodiments;

(19) FIG. 18 is a flow diagram of a method for assembling a flexible display screen, in accordance with some embodiments;

(20) FIG. 19 is a structural comparison diagram of a flexible display screen assembled and unassembled through carrier films, in accordance with some embodiments;

(21) FIG. 20 is a sectional view taken along the FF' direction in FIG. 19;

(22) FIG. 21 is a structural diagram showing an assembly of a flexible display screen, in accordance with some embodiments;

(23) FIG. 22 is a simulation diagram showing variation and distribution of stress in a case where a flexible display screen is not provided with openings, in accordance with some embodiments; and

(24) FIG. 23 is a simulation diagram showing variation and distribution of stress in a case where a flexible display screen is provided with openings, in accordance with some embodiments.

DETAILED DESCRIPTION

(25) Technical solutions in some embodiments of the present disclosure will be described clearly and completely below with reference to the accompanying drawings. Obviously, the described embodiments are merely some but not all embodiments of the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present disclosure shall be included in the protection scope of the present disclosure.

(26) Unless the context requires otherwise, throughout the description and the claims, the term "comprise" and other forms thereof such as the third-person singular form "comprises" and the present participle form "comprising" are construed as an open and inclusive meaning, i.e., "including, but not limited to." In the description of the specification, the terms such as "one embodiment," "some embodiments," "exemplary embodiments," "an example," "specific example" or "some examples" are intended to indicate that specific features, structures, materials or characteristics related to the embodiment(s) or example(s) are included in at least one embodiment or example of the present disclosure. Schematic representations of the above terms do not necessarily refer to the same embodiment(s) or example(s). In addition, the specific features, structures, materials or characteristics may be included in any one or more embodiments or examples in any suitable manner.

(27) The phrase "A and/or B" includes following three combinations: only A, only B, and a combination of A and B.

(28) The term such as "about," "substantially" or "approximately" as used herein includes a stated value and an average value within an acceptable range of deviation of a particular value determined by a person of ordinary skill in the art, considering measurement in question and errors associated with measurement of a particular quantity (i.e., limitations of a measurement system).

(29) Exemplary embodiments are described herein with reference to sectional views and/or plan views as idealized exemplary drawings. In the accompanying drawings, thicknesses of layers and sizes of regions are enlarged for clarity. Thus, variations in shape relative to the accompanying drawings due to, for example, manufacturing techniques and/or tolerances may be envisaged. Therefore, the exemplary embodiments should not be construed to be limited to the shapes of regions shown herein, but to include deviations in shape due to, for example, manufacturing. For example, an etched region shown in a rectangular shape generally has a curved feature. Therefore, the regions shown in the accompanying drawings are schematic in nature, and their shapes are not intended to show actual shapes of the regions in a device, and are not intended to limit the scope of the exemplary embodiments.

(30) In the description of the present disclosure, it will be understood that orientations or positional relationships indicated by the terms such as "center," "longitudinal," "lateral," "length," "width,"

“vertical,” “horizontal,” “inner” and “outer” are based on orientations or positional relationships shown in the drawings, which are merely to facilitate the description of the present disclosure and simplify the description, and are not to indicate or imply that the indicated devices or elements must have particular orientations, or must be constructed or operated in particular orientations, and thus cannot be construed as limitations on the present disclosure.

(31) In addition, a feature defined with “first” or “second” may explicitly or implicitly include one or more of the features. In the description of the embodiments of the present disclosure, the term “a plurality of/the plurality of” means two or more unless otherwise specified.

(32) In the description of the present disclosure, it will be noted that the terms such as “installed,” “joined” and “connected” shall be construed broadly, unless otherwise explicitly stated or limited. For example, the term “connected” may be fixedly connected, detachably connected, or integrally connected; the term “connected” may be mechanically connected or electrically connected; the term “connected” may be directly connected, indirectly connected through an intermediate medium, or internally communicated between two elements. Specific meanings of the above terms in the present disclosure may be understood as appropriate by a person of ordinary skill in the art.

(33) As shown in FIG. 1, some embodiments of the present disclosure provide an electronic apparatus. The electronic apparatus **101** may be a television, a mobile phone, a computer, a notebook computer, a tablet computer, a personal digital assistant (PDA), or a vehicle-mounted computer.

(34) The electronic apparatus **101** includes a display panel. The display panel may be a liquid crystal display (LCD) panel. Alternatively, the display panel may be an electroluminescent display panel or a photoluminescent display panel. In a case where the display panel is the electroluminescent display panel, the electroluminescent display panel may be an organic light-emitting diode (OLED) display panel or a quantum dot light-emitting diode (QLED) display panel. In a case where the display panel is the photoluminescent display panel, the photoluminescent display panel may be a quantum dot photoluminescent display panel.

(35) The display panel includes a flexible display screen **102** and a forming die **103**. In order to obtain a curved electronic apparatus, the flexible display screen **102** of the display panel is usually arranged to be curved and attached to the forming die **103**. For example, an edge of the forming die **103** is processed to have a curved surface, and in a process of attaching the flexible display screen **102** to the forming die **103**, an edge of the flexible display screen **102** is curved and attached to the forming die **103**.

(36) As shown in FIG. 1, the flexible display screen **102** includes a main display portion **1** and an edge display portion **2**. The edge display portion **2** is arranged around an edge of the main display portion **1**, and is fixedly connected to the main display portion **1** or integrally formed with the main display portion **1**. As shown in FIG. 2, the edge display portion **2** is curved toward the forming die **103**, and is attached to the forming die **103**.

(37) In embodiments, the main display portion **1** may be located in the same plane. That is, the main display portion **1** has a planar structure. Of course, the main display portion **1** may be an arc-shaped display portion. That is, heights of two opposite ends of the main display portion **1** are slightly less than a height of the middle of the main display portion **1**. The main display portion **1** includes a first side **11** and a second side **12** that are adjacent to each other. The first side **11** is connected to the second side **12** through an arc-shaped side **13**, so that an outer contour of the main display portion **1** may be smooth.

(38) As shown in FIG. 2, the edge display portion **2** includes a first bending portion **21**, a second bending portion **22** and a corner bending portion **23**. The first bending portion **21**, the corner bending portion **23** and the second bending portion **22** are connected in sequence. The first bending portion **21** is located on a side of the main display portion **1** where the first side **11** is located, and is connected to the first side **11**. The second bending portion **22** is located on a side of the main display portion **1** where the second side **12** is located, and is connected to the second side **12**. The

corner bending portion **23** is located on a side of the main display portion **1** where the arc-shaped side **13** is located, and is connected to the arc-shaped side **13**.

(39) In some embodiments, a side of the first bending portion **21** is connected to the first side **11**, and after the first bending portion **21** is attached to the forming die **103**, the first bending portion **21** and the main display portion **1** may form an included angle therebetween. A side of the second bending portion **22** is connected to the second side **12**, and after the second bending portion **22** is attached to the forming die **103**, the second bending portion **22** and the main display portion **1** may form an included angle therebetween. A side of the corner bending portion **23** is connected to the arc-shaped side **13**, and after the corner bending portion **23** is attached to the forming die **103**, the corner bending portion **23** and the main display portion **1** may form an included angle therebetween.

(40) After the flexible display screen **102** is attached to the forming die **103**, the first bending portion **21**, the second bending portion **22** and the corner bending portion **23** have the same bending direction B. The first bending portion **21**, the corner bending portion **23** and the second bending portion **22** are connected in sequence, and constitute a display region AA together with the main display portion **1**. The flexible display screen **102** with a curved edge has a large display area, and the edge is smooth.

(41) In some embodiments, the first side **11** and the second side **12** are straight sides, so that the structure of the flexible display screen **102** may be simplified, and the flexible display screen **102** and the forming die **103** may be attached conveniently.

(42) In some embodiments, the main display portion **1** includes first sides **11** and second sides **12**, and any two adjacent sides respectively a first side **11** and a second side **12**, and are connected through an arc-shaped side **13**. Each arc-shaped side **13** is connected to a corner bending portion **23**, each first side **11** is connected to a first bending portion **21**, and each second side **12** is connected to a second bending portion **22**. A plurality of corner bending portions **23**, first bending portions **21** and second bending portions **22** are connected, and constitute the display region AA of the flexible display screen **102** together with the main display portion **1**. That is, the corner bending portions **23**, the first bending portions **21** and the second bending portions **22** are each provided with pixels to realize a display function.

(43) For the flexible display screen **102** and the forming die **103** in the attaching process of the flexible display screen **102**, a corner of the forming die **103** generally has a spherical-like surface, so that the flexible display screen **102** is subjected to a large tensile stress along a radial direction (i.e., a radial direction of the arc-shaped side **13**, Y direction as shown in FIG. 3) and a large tensile stress along a lateral direction (i.e., an extending direction of the arc-shaped side **13**, X direction as shown in FIG. 3) at the middle of the corner. Therefore, the middle of the corner bending portion **23** generates a larger tensile stress than the first bending portion **21** or the second bending portion **22**, which is more likely to cause cracks, wrinkles or attachment cracking, resulting in a damage to the screen.

(44) As shown in FIG. 3, in some embodiments, a plurality of openings **4** are provided in the corner bending portion, and the plurality of openings **4** face the bending direction B of the corner bending portion. In a bending process of the corner bending portion **23**, the tensile stress to which the corner bending portion **23** is subjected is partially absorbed by deformation of the openings **4**, so as to reduce a risk of cracking and wrinkling of the corner bending portion **23**.

(45) As shown in FIG. 3, in exemplary embodiments, the corner bending portion **23** has a wiring region **231** and a deformation region **232**. The deformation region **232** is close to the main display portion **1**, and is connected to the main display portion **1**. The wiring region **231** is located on a side of the deformation region **232** away from the main display portion **1**.

(46) The wiring region is used for arranging one or more of some components, pixel circuits and signal line(s).

(47) For example, as shown in FIG. 3, the wiring region is provided with first signal line(s) M1

therein.

(48) The wiring region **231** is provided with at least two first signal line **M1** therein.

(49) The first signal line **M1** may be, for example, a data line for transmitting a gray-scale data signal to a pixel, or a supply voltage signal line.

(50) In some embodiments, the deformation region **232** is provided with the plurality of openings **4** therein. The tensile stress to which the corner bending portion **23** is subjected is partially absorbed by the deformation of the openings **4**, which reduces the risk of cracking and wrinkling of the corner bending portion **23**, and reduces a stress strength of the wiring region **231**, thereby protecting structures such as the components and the pixel circuits in the wiring region **231**.

(51) It will be understood that the wiring region **231** is not provided with an opening **4**, which improves a strength of the wiring region **231**, and makes the wiring region **231** difficult to deform relative to the deformation region **232**, so that the signal line(s) in the wiring region **231** are protected to reduce a risk of cracking.

(52) In some embodiments, adjacent openings **4** are provided with at least one sub-pixel **5** there between. Moreover, gaps between adjacent openings **4** may be used for arranging signal lines to realize the display function.

(53) The openings **4** may be reasonably distributed according to distances between adjacent sub-pixels **5**.

(54) In embodiments, the corner bending portion **23** further includes a plurality of second signal lines **M2**.

(55) In embodiments, in a case where the corner bending portion **23** has the wiring region **231** and the deformation region **232**, the second signal lines **M2** are disposed in the deformation region **232**.

(56) Each sub-pixel **5** is electrically connected to at least one second signal line **M2**.

(57) The plurality of second signal lines **M2** extend in gaps between openings **4**, and each second signal line **M2** is electrically connected to a first signal line **M1**.

(58) The second signal line **M2** may be electrically connected to at least one sub-pixel **5**. The second signal line **M2** may be, for example, a data line for transmitting a gray-scale data signal to a pixel, or a supply voltage signal line.

(59) The second signal line **M2** may be arranged into a polygonal line or a curved line to enhance stretchability.

(60) In some embodiments, as shown in FIG. 3, the plurality of openings **4** are arranged in a plurality of rows along the radial direction (i.e., the Y direction) of the arc-shaped side **13**, and openings **4** in each row are arranged at intervals along the extending direction (i.e., the X direction) of the arc-shaped side **13**. Thus, when the flexible display screen **102** is attached to the forming die **103**, the stress and strain may be better absorbed, the problem of the attachment cracking of the flexible display screen **102** may be better solved, and light-emitting devices and pixel circuits in a region without openings **4** may be better protected.

(61) In exemplary embodiments, as shown in FIG. 4, along a direction (i.e., X1 direction) from a center of the deformation region **232** toward the first bending portion **21** and along a direction (i.e., X2 direction) from the center of the deformation region **232** toward the second bending portion **22**, dimensions of openings **4** in each of at least one row along the radial direction (i.e., Y direction) of the arc-shaped side **13** are gradually reduced. After the corner bending portion **23** is curved and attached to the forming die **103**, a center of a region where openings **4** in a row are located is stressed more than two ends of the region. Therefore, dimensions of the openings **4** in this row along the radial direction of the arc-shaped side **13** are gradually reduced from the center to the two ends, so that the center of the region where the openings **4** in this row are located absorbs more stress. Thus, the region where the openings **4** in this row are located is uniformly stressed along the radial direction of the arc-shaped side **13** after the attaching.

(62) In exemplary embodiments, as shown in FIG. 5, along the direction (i.e., X1 direction) from the center of the deformation region **232** toward the first bending portion **21** and along the direction

(i.e., X2 direction) from the center of the deformation region **232** toward the second bending portion **22**, dimensions of openings **4** in each of at least one row along the extending direction (i.e., X direction) of the arc-shaped side **13** are gradually reduced. From a center of a region where openings **4** in a row are located to two ends of the region, dimensions of the openings **4** in this row along the extending direction of the arc-shaped side **13** are gradually reduced, so that the center of the region where the openings **4** in this row are located absorbs more stress. Thus, the region where the openings **4** in this row are located is uniformly stressed along the extending direction of the arc-shaped side **13** after the attaching.

(63) In exemplary embodiments, as shown in FIG. 6, along the direction (i.e., X1 direction) from the center of the deformation region **232** toward the first bending portion **21** and along the direction (i.e., X2 direction) from the center of the deformation region **232** toward the second bending portion **22**, openings **4** in each of at least one row are gradually reduced in dimension along the radial direction (i.e., Y direction) of the arc-shaped side **13**, and are gradually reduced in dimension along the extending direction (i.e., X direction) of the arc-shaped side **13**. The dimensions of the openings **4** along the radial direction of the arc-shaped side **13** and the extending direction of the arc-shaped side **13** are designed, so that the dimensions of the openings **4** are suitable for the stress to which the corner bending portion **23** is subjected. That is, openings **4** in a center of a region where openings **4** in a row are located that are stressed more have respective large dimensions and absorb more stress, and openings **4** in two ends of the region that are stressed less have respective small dimensions and absorb less stress. Thus, the region where the openings **4** in this row are located is uniformly stressed along the radial direction of the arc-shaped side **13** and the extending direction of the arc-shaped side **13** after the attaching, so that the risk of damage to the corner bending portion **23** is effectively reduced.

(64) In exemplary embodiments, as shown in FIG. 7, along a direction (i.e., Y1 direction) from the center of the deformation region **232** toward a side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13** and along a direction (i.e., Y2 direction) from the center of the deformation region **232** toward another side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13**, dimensions, along the radial direction (i.e., Y direction) of the arc-shaped side **13**, of openings **4** arranged along the radial direction of the arc-shaped side **13** are gradually reduced. After the corner bending portion **23** is curved and attached to the forming die **103**, a center of a region where the openings **4** are located is stressed more than two ends of the region. Therefore, the dimensions of the openings **4** along the radial direction of the arc-shaped side **13** are gradually reduced from the center to the two ends, so that the center of the region where the openings **4** are located absorbs more stress. Thus, the region where the openings **4** are located is uniformly stressed along the radial direction of the arc-shaped side **13** after the attaching.

(65) In exemplary embodiments, as shown in FIG. 8, along the direction (i.e., Y1 direction) from the center of the deformation region **232** toward the side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13** and along the direction (i.e., Y2 direction) from the center of the deformation region **232** toward the another side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13**, dimensions, along the extending direction (i.e., X direction) of the arc-shaped side **13**, of openings **4** arranged along the radial direction of the arc-shaped side **13** are gradually reduced. From a center of a region where the openings **4** are located to two ends of the region, the dimensions of the openings **4** along the extending direction of the arc-shaped side **13** are gradually reduced, so that the center of the region where the openings **4** are located absorbs more stress. Thus, the region where the openings **4** are located is uniformly stressed along the extending direction of the arc-shaped side **13** after the attaching.

(66) In exemplary embodiments, as shown in FIG. 9, along the direction (i.e., Y1 direction) from the center of the deformation region **232** toward the side boundary of the deformation region **232**

along the radial direction of the arc-shaped side **13** and along the direction (i.e., Y2 direction) from the center of the deformation region **232** toward the another side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13**, openings **4** arranged along the radial direction of the arc-shaped side **13** are gradually reduced in dimension along the radial direction of the arc-shaped side **13**, and are gradually reduced in dimension along the extending direction of the arc-shaped side **13**. The dimensions of the openings **4** along the radial direction of the arc-shaped side **13** and the extending direction of the arc-shaped side **13** are designed, so that the dimensions of the openings **4** are suitable for the stress to which the corner bending portion **23** is subjected. That is, openings **4** in a center of a region where the openings **4** are located that are stressed more have respective large dimensions and absorb more stress, and openings **4** in two ends of the region that are stressed less have respective small dimensions and absorb less stress. Thus, the region where the openings **4** are located is uniformly stressed along the radial direction of the arc-shaped side **13** and the extending direction of the arc-shaped side **13** after the attaching, so that the risk of damage to the corner bending portion **23** is effectively reduced.

(67) In exemplary embodiments, as shown in FIG. **10**, along the direction (i.e., X1 direction) from the center of the deformation region **232** toward the first bending portion **21** and along the direction (i.e., X2 direction) from the center of the deformation region **232** toward the second bending portion **22**, openings **4** in each row are gradually reduced in dimension along the extending direction of the arc-shaped side **13**, and are gradually reduced in dimension along the radial direction of the arc-shaped side **13**; moreover, along the direction (i.e., Y1 direction) from the center of the deformation region **232** toward the side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13** and along the direction (i.e., Y2 direction) from the center of the deformation region **232** toward the another side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13**, openings **4** arranged along the radial direction of the arc-shaped side are gradually reduced in dimension along the extending direction of the arc-shaped side **13**, and are gradually reduced in dimension along the radial direction of the arc-shaped side **13**. Thus, after the corner bending portion **23** is attached to the forming die **103**, the corner bending portion **23** is uniformly stressed along the X direction and the Y direction.

(68) In exemplary embodiments, as shown in FIG. **11**, the deformation region **232** includes a middle region **233** and an edge region **234**, and the edge region **234** is arranged around the middle region **233**.

(69) In embodiments, a dimension of an opening **4** in the middle region **233** along the radial direction (i.e., Y direction in FIG. **11**) of the arc-shaped side **13** is greater than a dimension of an opening **4** in the edge region **234** along the radial direction of the arc-shaped side **13**. After the corner bending portion **23** is curved and attached to the forming die **103**, the middle region **233** is stressed more than the edge region **234**. Therefore, compared to the edge region **234**, the dimension of opening **4** in the middle region **233** is larger, so that the middle region **233** absorbs more stress. Thus, the middle region **233** and the edge region **234** are uniformly stressed along the radial direction of the arc-shaped side **13** after the attaching.

(70) In embodiments, a dimension of an opening **4** in the middle region **233** along the extending direction (i.e., X direction in FIG. **11**) of the arc-shaped side **13** is greater than a dimension of an opening **4** in the edge region **234** along the extending direction of the arc-shaped side **13**. Thus, the middle region **233** and the edge region **234** are uniformly stressed along the extending direction of the arc-shaped side **13** after the attaching.

(71) In embodiments, a dimension of an opening **4** in the middle region **233** along the radial direction (i.e., Y direction in FIG. **11**) of the arc-shaped side **13** is greater than a dimension of an opening **4** in the edge region **234** along the radial direction of the arc-shaped side **13**, and a dimension of the opening **4** in the middle region **233** along the extending direction of the arc-shaped side **13** is greater than a dimension of the opening **4** in the edge region **234** along the extending direction of the arc-shaped side **13**. Thus, the middle region **233** and the edge region **234**

are uniformly stressed along the radial direction of the arc-shaped side **13** and the extending direction of the arc-shaped side **13** after the attaching.

(72) In embodiments, dimensions of openings **4** in the middle region **233** along the radial direction (i.e., Y direction in FIG. **11**) of the arc-shaped side **13** are substantially the same. Thus, the structure of the corner bending portion **23** may be simplified, so as to simplify the structure of the flexible display screen **102**, which may improve the production efficiency of the flexible display screen **102**.

(73) In embodiments, dimension of openings **4** in the edge region **234** along the radial direction (i.e., Y direction in FIG. **11**) of the arc-shaped side **13** are substantially the same. Thus, the structure of the corner bending portion **23** may be simplified, so as to simplify the structure of the flexible display screen **102**, which may improve the production efficiency of the flexible display screen **102**.

(74) In embodiments, dimensions of the openings **4** in the middle region **233** along the extending direction (i.e., X direction in FIG. **11**) of the arc-shaped side **13** are substantially the same. Thus, the structure of the corner bending portion **23** may be simplified, so as to simplify the structure of the flexible display screen **102**, which may improve the production efficiency of the flexible display screen **102**.

(75) In embodiments, dimensions of the openings **4** in the edge region **234** along the extending direction (i.e., X direction in FIG. **11**) of the arc-shaped side **13** are substantially the same. Thus, the structure of the corner bending portion **23** may be simplified, so as to simplify the structure of the flexible display screen **102**, which may improve the production efficiency of the flexible display screen **102**.

(76) In embodiments, dimensions of the openings **4** in the middle region **233** along the radial direction (i.e., Y direction in FIG. **11**) of the arc-shaped side **13** and the extending direction (i.e., X direction in FIG. **11**) of the arc-shaped side **13** are substantially the same, and dimensions of the openings **4** in the edge region **234** along the radial direction of the arc-shaped side **13** and the extending direction of the arc-shaped side **13** are substantially the same. Thus, the structure of the corner bending portion **23** may be simplified, so as to simplify the structure of the flexible display screen **102**, which may improve the production efficiency of the flexible display screen **102**.

(77) In exemplary embodiments, as shown in FIG. **11**, a dimension of the deformation region **232** along the radial direction (i.e., Y direction) of the arc-shaped side **13** is L1. A minimum distance d1, along the radial direction of the arc-shaped side **13**, between a boundary of the middle region **233** and a boundary of the deformation region **232** that are close to each other is in a range of $\frac{1}{4}$ L1 to $\frac{1}{3}$ L1. For example, the minimum distance d1 is 0.25 L1, 0.26 L1, 0.275 L1, 0.3 L1 or 0.33 L1.

(78) In exemplary embodiments, as shown in FIG. **11**, a dimension of the deformation region **232** along the extending direction (i.e., X direction) of the arc-shaped side **13** is L2. A distance d2, along the extending direction of the arc-shaped side **13**, between a boundary of the middle region **233** and a boundary of the deformation region **232** that are close to each other is in a range of $\frac{1}{6}$ L2 to $\frac{1}{5}$ L2. For example, the distance d2 is 0.166 L2, 0.17 L2, 0.175 L2, 0.18 L2 or 0.2 L2.

(79) It will be noted that L2 and d2 in FIG. **11** are not linear lengths, L2 is a total arc length, along the X direction, of a region where openings **4** in a row are located, and d2 is an arc length, along the X direction, between a portion of the boundary of the middle region **233** in the region where the openings **4** in this row are located and a portion of the boundary of the deformation region **232** in the region where the openings **4** in this row are located.

(80) In exemplary embodiments, as shown in FIG. **12**, along the direction (i.e., X1 direction) from the center of the deformation region **232** toward the first bending portion **21** and along the direction (i.e., X2 direction) from the center of the deformation region **232** toward the second bending portion **22**, distances, each of which is a distance between two adjacent openings **4** in openings **4** in each of at least one row, are gradually increased.

(81) In exemplary embodiments, as shown in FIG. **13**, along the direction (i.e., Y1 direction) from

the center of the deformation region **232** toward the side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13** and along the direction (i.e., Y2 direction) from the center of the deformation region **232** toward the another side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13**, distances, each of which is a distance between two adjacent openings **4** in openings **4** arranged along the radial direction of the arc-shaped side **13**, are gradually increased.

(82) In embodiments, along the direction from the center of the deformation region **232** toward the first bending portion **21** and along the direction from the center of the deformation region **232** toward the second bending portion **22**, distances, each of which is a distance between two adjacent openings **4** in openings **4** in each of at least one row, are gradually increased. Moreover, along the direction from the center of the deformation region **232** toward the side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13** and along the direction from the center of the deformation region **232** toward the another side boundary of the deformation region **232** along the radial direction of the arc-shaped side **13**, distances, each of which is a distance between two adjacent openings **4** in openings **4** arranged along the radial direction of the arc-shaped side **13**, are gradually increased. Therefore, from the center of the deformation region **232** to four sides of the corner bending portion **23**, the stress to which the corner bending portion **23** is subjected gradually decreases. Thus, from the center of the deformation region **232** to the four sides of the corner bending portion **23**, an arrangement density of the openings **4** is gradually reduced, so that a whole of the corner bending portion **23** is uniformly stressed.

(83) In exemplary embodiments, as shown in FIG. **14**, the plurality of openings **4** are divided into a plurality of groups, and the plurality of groups of openings **4** are arranged in a plurality of rows along the radial direction (i.e., Y direction) of the arc-shaped side **13**. Groups of openings **4** in each row are arranged at intervals along the extending direction (i.e., X direction) of the arc-shaped side **13** or a tangential direction Z of the arc-shaped side **13**. A distance between two adjacent openings **4** in each group of openings **4** is less than a distance between two openings **4** that are closest to each other in two adjacent groups of openings **4**. Thus, the structural diversity of the opening **4** may be increased, which is beneficial to improving the structural strength of the flexible display screen **102**.

(84) In embodiments, the number of openings **4** constituting each group of openings **4** in the middle region **233** is greater than the number of openings **4** constituting each group of openings **4** in the edge region **234**. Two adjacent groups of openings are provided with at least one sub-pixel therebetween.

(85) In exemplary embodiments, as shown in FIG. **15**, the opening **4** is a through hole penetrating through the corner bending portion **23** along a thickness direction T of the corner bending portion **23**. The opening **4** arranged in the through hole is able to absorb more stress, so that the corner bending portion **23** is ensured to be uniformly stressed to a great extent.

(86) In exemplary embodiments, as shown in FIG. **16**, the opening **4** is a groove recessed along the thickness direction T of the corner bending portion **23**. A notch of the groove faces the bending direction B of the corner bending portion **23**. The groove-shaped opening **4** may ensure the structural strength on a premise of absorbing the stress.

(87) In exemplary embodiments, as shown in FIG. **17**, the openings **4** may have one or more of a circular shape (referring to FIG. **14**), an elliptical shape (referring to (IV) in FIG. **17**), or a polygonal shape (referring to (I), (II), (III) or (V) in FIG. **17**). Thus, the structural diversity of the opening **4** may be increased. For example, the opening **4** has a hexagonal shape (referring to (I) in FIG. **17**), and has a narrow middle and two wide ends. For example, the opening **4** has a trapezoidal shape (referring to (II) in FIG. **17**). For example, the opening **4** has a strip shape. A length extending direction of the opening **4** is substantially the same as the extending direction of the arc-shaped side (referring to (V) in FIG. **17**); and/or a length extending direction of the opening **4** is substantially tangent to the extending direction of the arc-shaped side (referring to (III) in FIG.

17).

(88) In embodiments, the plurality of openings **4** have the same shape. Thus, the structure of the corner bending portion **23** may be simplified, so as to simplify the structure of the flexible display screen **102**. For example, each opening **4** has a circular shape or a rectangular shape.

(89) As shown in FIG. **18**, embodiments of the present disclosure provide a method for assembling the flexible display screen **102**.

(90) In some embodiments, as shown in FIG. **18**, the method for assembling the flexible display screen **102** includes **S1** to **S3**.

(91) In **S1**, as shown in FIG. **21**, a side of the first bending portion **21**, a side of the second bending portion **22**, and a side of the corner bending portion **23** that are away from the main display portion **1** are connected to respective carrier films **3**.

(92) In **S2**, the flexible display screen **102** is attached to the forming die **103**. At least part of each of the carrier films protrudes from a boundary of the forming die.

(93) In **S3**, the carrier films **3** are pulled to completely attach the flexible display screen **102** to the forming die **103**.

(94) Through the assembling method, the edge of the flexible display screen **102** reaches an optimal attaching position of the forming die **103**. In this case, the whole of the corner bending portion **23** is uniformly stressed, and an edge of the corner bending portion **23** away from the arc-shaped side **13** of the main display portion **1** is neither in tension nor in compression, which reduces the risk of wrinkling and cracking of the corner bending portion **23**. In a case where the corner bending portion **23** of the flexible display screen **102** has the deformation region **232** and the wiring region **231**, the wiring region **231** may be effectively protected.

(95) As shown in FIG. **19**, in a case where the carrier film **3** is not used for pulling and assembling, the edge of the flexible display screen **102** can only reach a position indicated by the dotted line. After the carrier films **3** are used for assembling, a region enclosed by the edge of the flexible display screen **102** is expanded by a circle compared to the position of the dotted line.

(96) FIG. **20** is a sectional view taken along the FF' direction in FIG. **19**. In a case where the carrier film **3** is not used for assembling (e.g., in the bent state **1** shown in FIG. **20**), an intersection point of a section line and a boundary of the flexible display screen **102** is a G point (as shown in FIGS. **19** and **20**). In a case where the carrier films **3** are used for assembling (e.g., in the bent state **2** shown in FIG. **20**), an intersection point of a section line and the boundary of the flexible display screen **102** is an H point (as shown in FIGS. **19** and **20**). An intersection point of a section line and a boundary of the main display portion **1** of the flexible display screen **102** is an N point (as shown in FIGS. **19** and **20**).

(97) As shown in FIG. **20**, Q is an outer edge position of the corner bending portion **23** in a case where the corner bending portion **23** is not bent (i.e., in an unbent state). QN is a length of the corner bending portion **23** in the case where the corner bending portion **23** is not bent. GN is an arc length of the corner bending portion **23** after being bent in a case where the corner bending portion **23** is not pulled by the carrier film **3**. HN is an arc length of the corner bending portion **23** after being pulled by the carrier films **3**. Taking a boundary W where the N point is located as an axis, R1 is a radius of a circle passing through the G point along a circumferential direction of the axis. Taking the boundary W where the N point is located as an axis, R2 is a radius of a circle passing through the H point along the circumferential direction of the axis.

(98) In a case where the carrier film **3** is not used for pulling, after the corner bending portion **23** is attached, an outer edge of the corner bending portion **23** can only reach the G point. In this case, the arc length GN is equal to the length QN, and the radius R1 of the circle where the G point is located is less than the length QN of the corner bending portion **23** in a natural state before the corner bending portion **23** is attached. Therefore, in this case, the outer edge of the corner bending portion **23** is compressed, and wrinkles are easily generated.

(99) However, in a case where the carrier films **3** are used for pulling, after the corner bending

portion **23** is attached, the outer edge of the corner bending portion **23** may reach the H point. In this case, the radius R2 of the circle where the H point is located is equal to the length QN of the corner bending portion **23** in the natural state before the corner bending portion **23** is attached. Therefore, in this case, the outer edge of the corner bending portion **23** is neither in tension nor in compression, which reduces the risk of wrinkling and cracking of the outer edge of the corner bending portion **23**.

(100) In some embodiments, as shown in FIG. **21**, a side of the carrier film **3** away from the main display portion **1** is connected to a clip **32**, so that the carrier film **3** is subjected to a uniform pulling force, and the flexible display screen **102** is pulled more effectively.

(101) In exemplary embodiments, as shown in FIG. **21**, the carrier film **3** connected to the corner bending portion **23** is provided with a plurality of slits **31** distributed at intervals. The plurality of slits **31** are spaced apart along the extending direction of the arc-shaped side **13**. Each slit **31** extends along the radial direction of the arc-shaped side **13**, and an end of each slit **31** extends to a side of the carrier film **3** connected to the corner bending portion **23**. Thus, the carrier film **3** connected to the corner bending portion **23** may be divided into a plurality of portions, and each of the plurality of portions is independently stressed. Therefore, the plurality of portions are not influenced by each other, so as to adapt to different tensile stresses required by the corner bending portion **23** at different positions, so that after being attached, the corner bending portion **23** is stressed in a balanced manner at all positions.

(102) The flexible display screen **102** described above is uniformly stressed after the attaching is completed, which may effectively reduce a probability of cracking, wrinkling and damage at corners of the flexible display screen **102**.

(103) As shown in FIG. **22**, in a case where the openings **4** are not provided, a central region of the corner bending portion **23** of the flexible display screen **102** is stressed more than a surrounding region. In the attaching process, the corner bending portion **23** is easily wrinkled, resulting in a poor attaching, and the corner bending portion **23** is also prone to cracking due to an uneven stress. In use, the central region stressed more is easily damaged after being collided.

(104) As shown in FIG. **23**, after the openings **4** are provided, the whole of the corner bending portion **23** of the flexible display screen **102** is stressed less and uniformly, so that the damage of the corner bending portion **23** is effectively avoided.

(105) In FIG. **22** or **23**, a lighter colored region is stressed more, and a darker colored region is stressed less.

(106) In another aspect, as shown in FIG. **1**, the embodiments of the present disclosure provide the electronic apparatus **101**. The electronic apparatus **101** includes the flexible display screen **102**.

(107) The electronic apparatus **101** may be any apparatus that displays images whether moving (e.g., videos) or stationary (e.g., still images). More specifically, it is anticipated that the embodiments may be implemented in, or associated with, a variety of electronic apparatuses. The variety of electronic apparatuses are, for example (but not limit to), mobile phones, wireless apparatuses, personal data assistants (PDAs), hand-held or portable computers, global positioning system (GPS) receivers/navigators, cameras, MP4 video players, camcorders, game consoles, watches, clocks, calculators, television monitors, flat panel displays, computer monitors, auto displays (e.g., odometer displays), navigators, cockpit controllers and/or displays, camera view displays (e.g., rear-view camera displays in vehicles), electronic photos, electronic billboards or signs, projectors, architectural structures, packaging and aesthetic structures (e.g., displays for displaying an image of a piece of jewelry).

(108) The foregoing descriptions are only specific implementations of the present disclosure, but the protection scope of the present disclosure is not limited thereto. Changes or replacements that any person skilled in the art could conceive of within the technical scope of the present disclosure shall be included in the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be subject to the protection scope of the claims.

Claims

1. A flexible display screen, comprising: a main display portion including a first side, an arc-shaped side and a second side that are connected in sequence; a first bending portion located on a side of the main display portion where the first side is located and connected to the first side; a second bending portion located on a side of the main display portion where the second side is located and connected to the second side; and a corner bending portion located on a side of the main display portion where the arc-shaped side is located and connected to the arc-shaped side; wherein the first bending portion, the corner bending portion and the second bending portion are connected in sequence, and constitute a display region together with the main display portion; and a plurality of openings are disposed in the corner bending portion, and face a bending direction of the corner bending portion; and the plurality of openings have a same shape, the plurality of openings are arranged in a plurality of rows in a radial direction of the arc-shaped side, openings in each row are arranged at intervals along an extending direction of the arc-shaped side, and the openings in each row are arranged in a same arc level along the arc-shaped side.
2. The flexible display screen according to claim 1, wherein the corner bending portion has a wiring region and a deformation region that are sequentially disposed along a direction from the corner bending portion toward the main display portion; wherein the deformation region is connected to the arc-shaped side, and the plurality of openings are disposed in the deformation region; and at least one first signal line is disposed in the wiring region.
3. The flexible display screen according to claim 2, wherein along a direction from a center of the deformation region toward the first bending portion and along a direction from the center of the deformation region toward the second bending portion, dimensions of openings in each of at least one row along the radial direction of the arc-shaped side are reduced in sequence; and/or dimensions of openings in each of at least one row along the extending direction of the arc-shaped side are reduced in sequence.
4. The flexible display screen according to claim 2, wherein along a direction from a center of the deformation region toward a side boundary of the deformation region along the radial direction of the arc-shaped side and along a direction from the center of the deformation region toward another side boundary of the deformation region along the radial direction of the arc-shaped side, dimensions, along the radial direction of the arc-shaped side, of openings arranged along the radial direction of the arc-shaped side are reduced in sequence; and/or dimensions, along the extending direction of the arc-shaped side, of openings arranged along the radial direction of the arc-shaped side are reduced in sequence.
5. The flexible display screen according to claim 2, wherein the deformation region includes a middle region and an edge region, and the edge region surrounds the middle region; wherein a dimension, along the radial direction of the arc-shaped side, of an opening in the plurality of openings located in the middle region is greater than a dimension, along the radial direction of the arc-shaped side, of an opening in the plurality of openings located in the edge region; and/or a dimension, along the extending direction of the arc-shaped side, of an opening in the plurality of openings located in the middle region is greater than a dimension, along the extending direction of the arc-shaped side, of an opening in the plurality of openings located in the edge region.
6. The flexible display screen according to claim 5, wherein dimensions, along the radial direction of the arc-shaped side, of openings in the plurality of openings located in the middle region are substantially same; and/or dimensions, along the radial direction of the arc-shaped side, of openings in the plurality of openings located in the edge region are substantially same.
7. The flexible display screen according to claim 5, wherein a dimension of the deformation region along the radial direction of the arc-shaped side is L1, and a minimum distance, along the radial direction of the arc-shaped side, between a boundary of the middle region and a boundary of the

deformation region that are close to each other is in a range of $\frac{1}{4} L1$ to $\frac{1}{3} L1$; and/or a dimension of the deformation region along the extending direction of the arc-shaped side is $L2$, and a distance, along the extending direction of the arc-shaped side, between a boundary of the middle region and a boundary of the deformation region that are close to each other is in a range of $\frac{1}{6} L2$ to $\frac{1}{8} L2$.

8. The flexible display screen according to claim 2, wherein along a direction from a center of the deformation region toward the first bending portion and along a direction from the center of the deformation region toward the second bending portion, distances, each of which is a distance between two adjacent openings in openings in each of at least one row, are increased in sequence; and/or along a direction from a center of the deformation region toward a side boundary of the deformation region along the radial direction of the arc-shaped side and along a direction from the center of the deformation region toward another side boundary of the deformation region along the radial direction of the arc-shaped side, distances, each of which is a distance between two adjacent openings in openings arranged along the radial direction of the arc-shaped side, are increased in sequence.

9. The flexible display screen according to claim 1, wherein the plurality of openings have a strip shape, and a length extending direction of the plurality of openings is substantially same as the extending direction of the arc-shaped side; and/or a length extending direction of the plurality of openings is substantially tangent to the extending direction of the arc-shaped side.

10. The flexible display screen according to claim 1, wherein the plurality of openings are divided into a plurality of groups that are arranged in a plurality of rows along the radial direction of the arc-shaped side, and groups of openings in each row are arranged at intervals along the extending direction of the arc-shaped side or a tangential direction of the arc-shaped side; and a distance between two adjacent openings in each group of openings is less than a distance between two openings that are closest to each other in two adjacent groups of openings.

11. The flexible display screen according to claim 1, wherein the plurality of openings have a circular shape, an elliptical shape, or a polygonal shape.

12. The flexible display screen according to claim 1, wherein the plurality of openings are through holes penetrating through the corner bending portion along a thickness direction of the corner bending portion; or the plurality of openings are grooves recessed along a thickness direction of the corner bending portion.

13. The flexible display screen according to claim 1, wherein the corner bending portion includes: a plurality of sub-pixels, wherein two adjacent openings are provided with at least one sub-pixel therebetween; or the plurality of openings are divided into a plurality of groups, and two adjacent groups of openings are provided with at least one sub-pixel therebetween.

14. The flexible display screen according to claim 13, wherein the corner bending portion has a wiring region and a deformation region, the plurality of openings are disposed in the deformation region, and at least one first signal line is disposed in the wiring region; and the corner bending portion further includes: a plurality of second signal lines, wherein each sub-pixel is electrically connected to at least one second signal line; the plurality of second signal lines extend in gaps between the plurality of openings, and each second signal line is electrically connected to a first signal line in the at least one first signal line.

15. A method for assembling a flexible display screen, wherein the flexible display screen is the flexible display screen according to claim 1; the method comprises: connecting a side of the first bending portion, a side of the second bending portion, and a side of the corner bending portion that are away from the main display portion to respective carrier films; attaching the flexible display screen to a forming die, at least part of each of the carrier films protruding from a boundary of the forming die; and pulling the carrier films to attach the flexible display screen to the forming die.

16. The method for assembling the flexible display screen according to claim 15, wherein a carrier film connected to the corner bending portion of the flexible display screen is provided with a plurality of slits arranged at intervals, and the plurality of slits are arranged at intervals along an

extending direction of the arc-shaped side of the main display portion of the flexible display screen; each slit extends along a radial direction of the arc-shaped side, and an end of each slit extends to a side of the carrier film connected to the corner bending portion.

17. An electronic apparatus comprising the flexible display screen according to claim 1.

18. The flexible display screen according to claim 3, wherein along a direction from the center of the deformation region toward a side boundary of the deformation region along the radial direction of the arc-shaped side and along a direction from the center of the deformation region toward another side boundary of the deformation region along the radial direction of the arc-shaped side, dimensions, along the radial direction of the arc-shaped side, of openings arranged along the radial direction of the arc-shaped side are reduced in sequence; and/or dimensions, along the extending direction of the arc-shaped side, of openings arranged along the radial direction of the arc-shaped side are reduced in sequence.

19. The flexible display screen according to claim 3, wherein the deformation region includes a middle region and an edge region, and the edge region surrounds the middle region; wherein a dimension, along the radial direction of the arc-shaped side, of an opening in the plurality of openings located in the middle region is greater than a dimension, along the radial direction of the arc-shaped side, of an opening in the plurality of openings located in the edge region; and/or a dimension, along the extending direction of the arc-shaped side, of an opening in the plurality of openings located in the middle region is greater than a dimension, along the extending direction of the arc-shaped side, of an opening in the plurality of openings located in the edge region.
